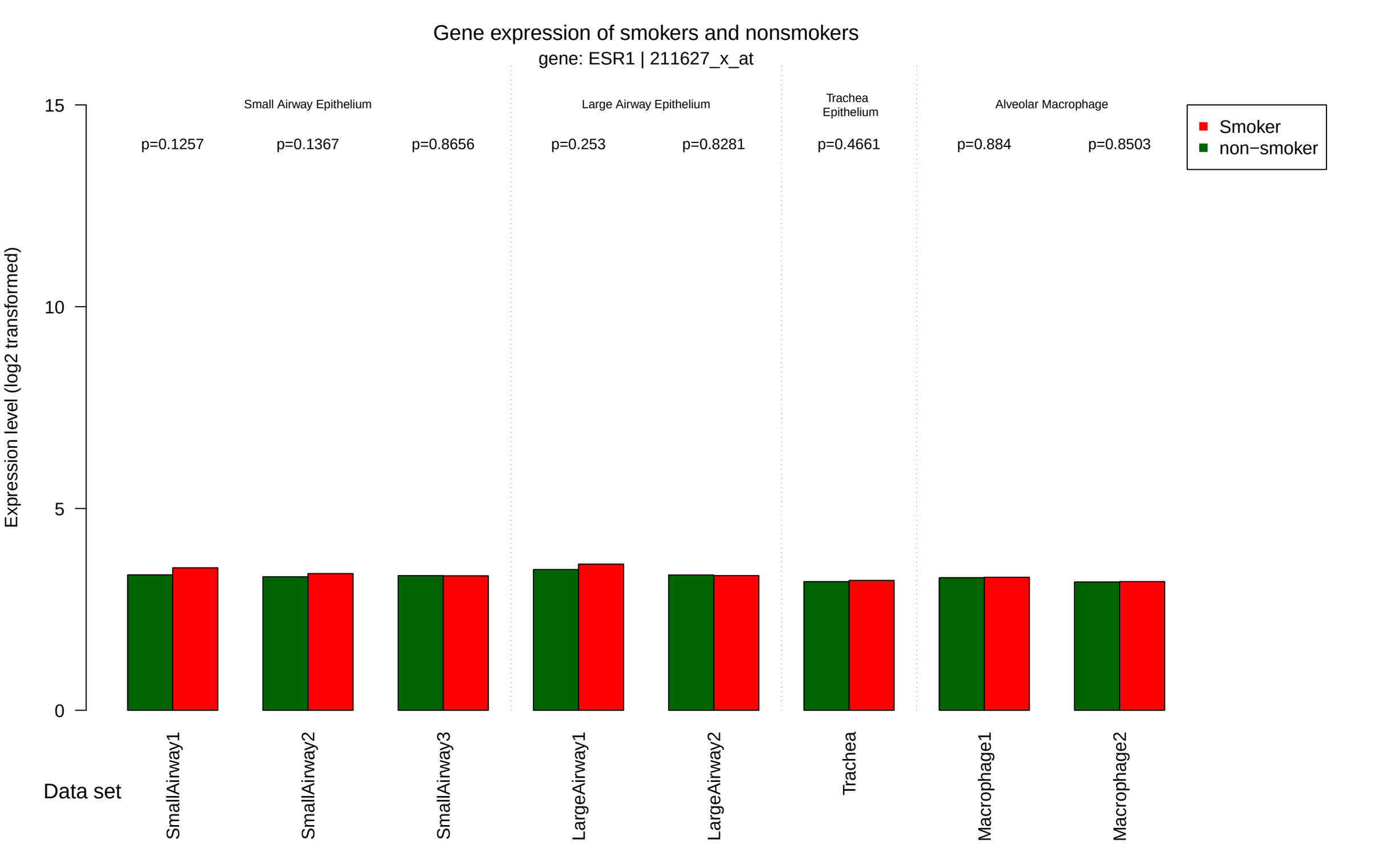
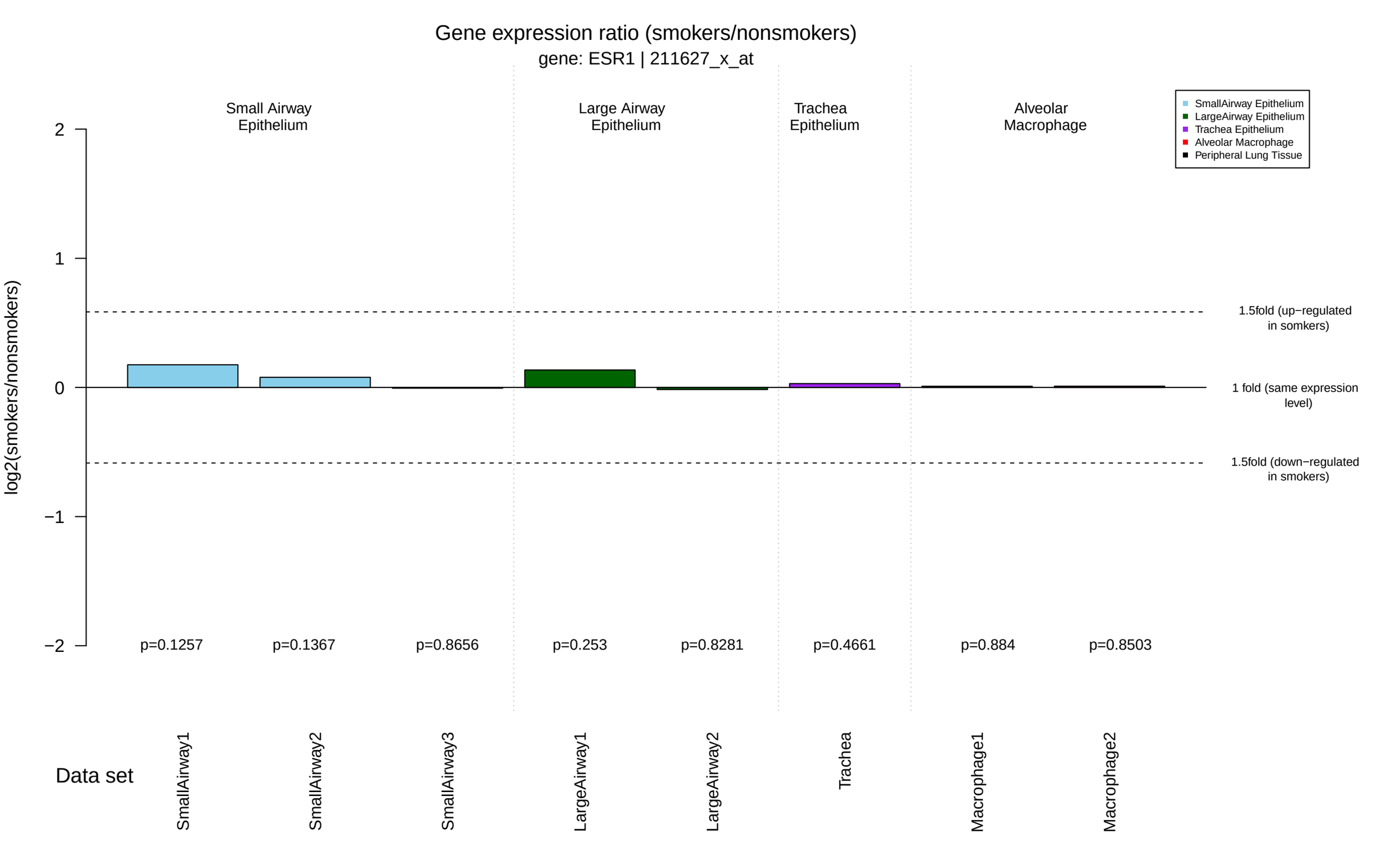
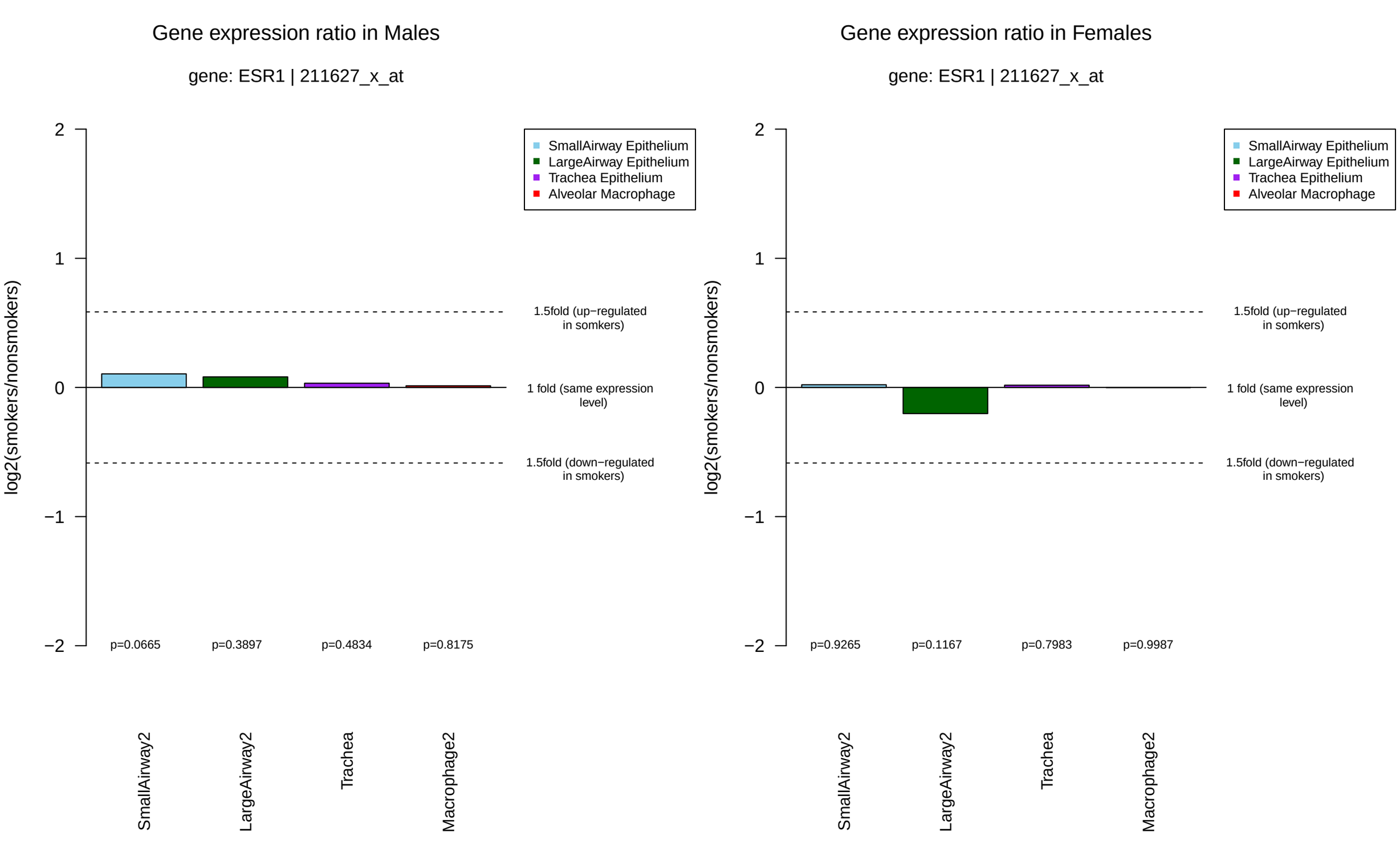
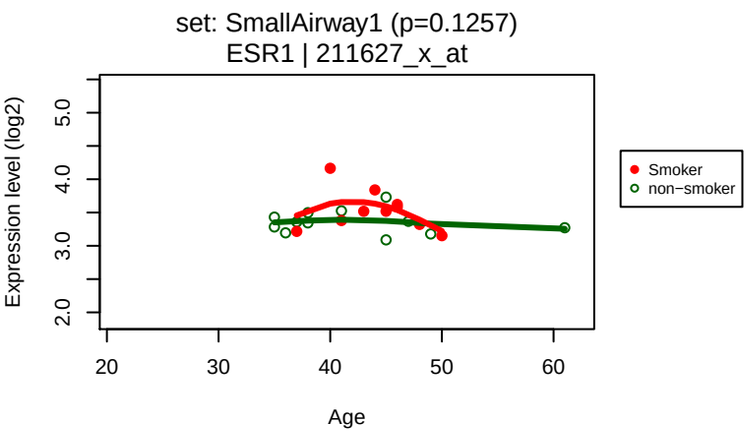
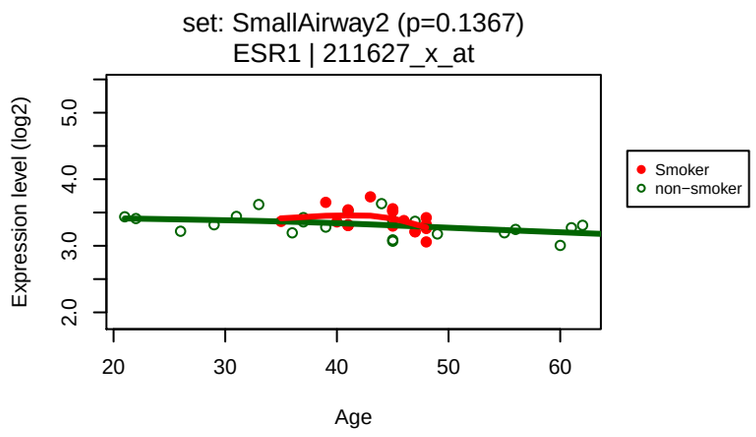
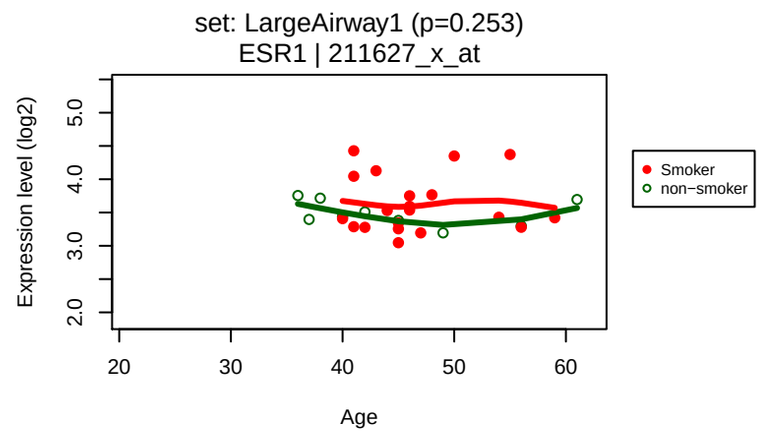

Gene expression of smokers and nonsmokers
gene: ESR1 | 211627_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1257
p=0.1367
p=0.8656
p=0.253
p=0.8281
p=0.4661
p=0.884
p=0.8503
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

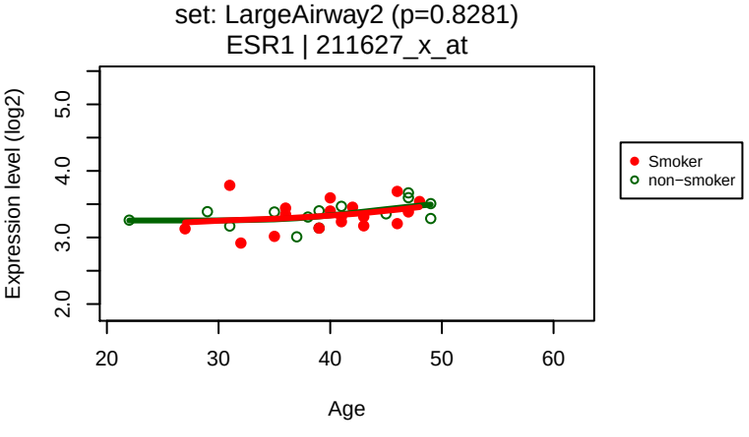

Gene expression ratio (smokers/nonsmokers)
gene: ESR1 | 211627_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1257
p=0.1367
p=0.8656
p=0.253
p=0.8281
p=0.4661
p=0.884
p=0.8503
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

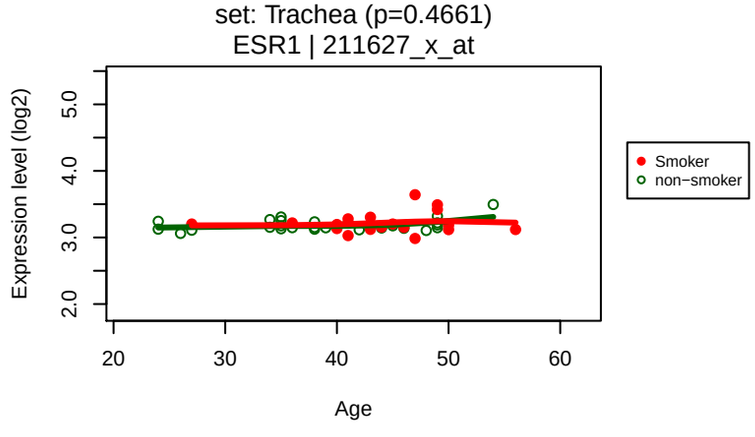

Gene expression ratio in Males
gene: ESR1 | 211627_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0665
p=0.3897
p=0.4834
p=0.8175
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ESR1 | 211627_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9265
p=0.1167
p=0.7983
p=0.9987
SmallAirway2
LargeAirway2
Trachea
Macrophage2

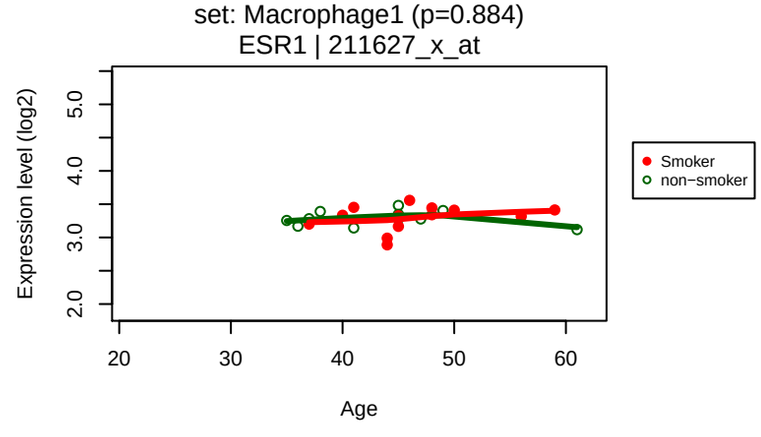

set: SmallAirway1 (p=0.1257)
ESR1 | 211627_x_at
Expression level (log2)
Age
Smoker
non−smoker

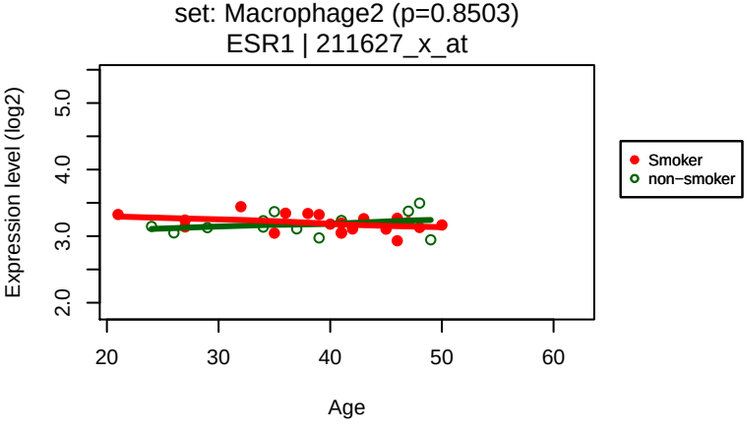

set: SmallAirway2 (p=0.1367)
ESR1 | 211627_x_at
Expression level (log2)
Age
Smoker
non−smoker

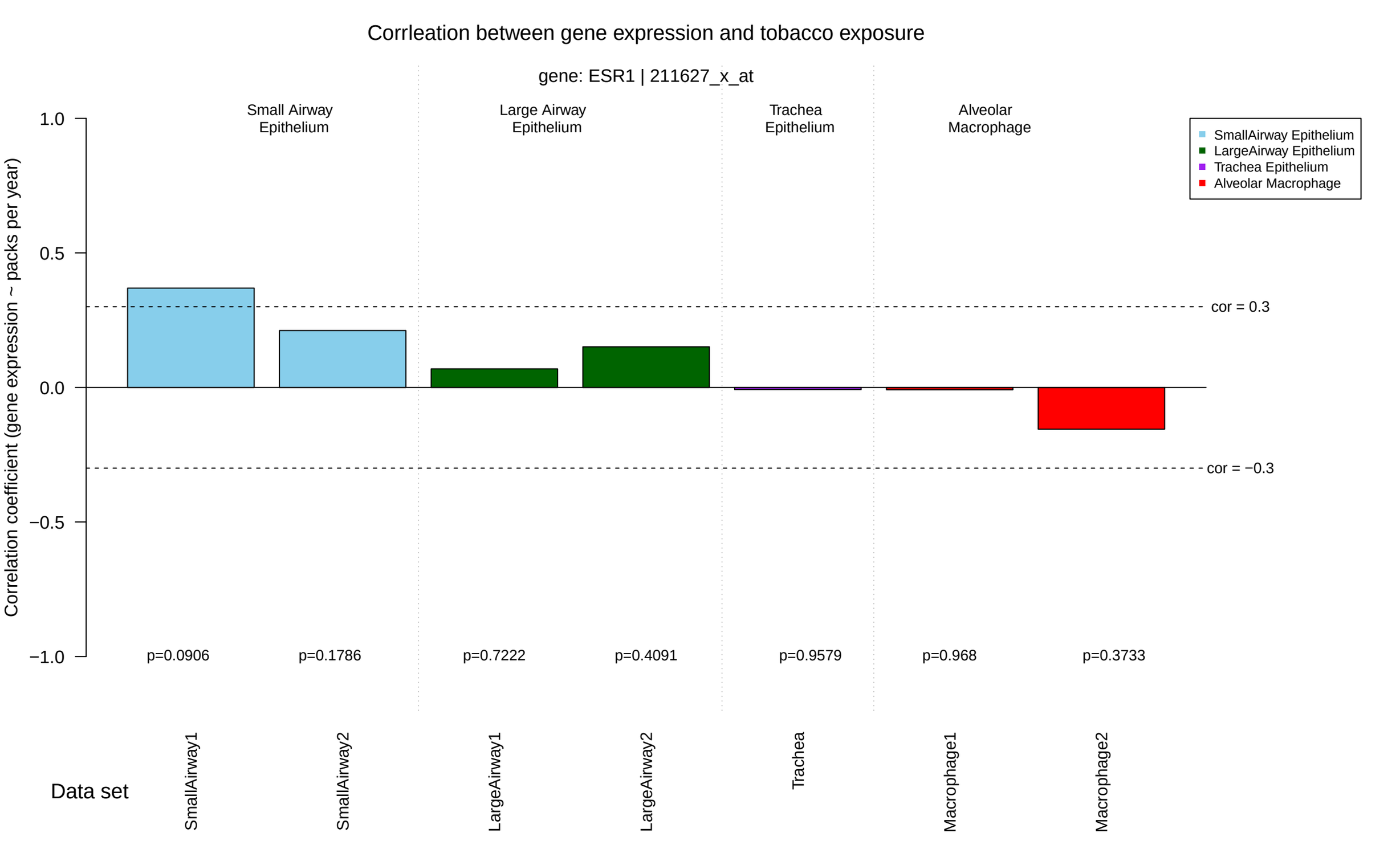

set: LargeAirway1 (p=0.253)
ESR1 | 211627_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8281)
ESR1 | 211627_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4661)
ESR1 | 211627_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.884)
ESR1 | 211627_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8503)
ESR1 | 211627_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ESR1 | 211627_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0906
p=0.1786
p=0.7222
p=0.4091
p=0.9579
p=0.968
p=0.3733
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2